



Shawn Arney

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Lead AI / Machine Learning Engineer / Architect

- 5 years **building and deploying** production-grade **ML solutions**.
- Proven track record in time series forecasting, end-to-end model development, and system integration using **Python, Docker, and AWS**.
- Proficient in implementing **CNN, LSTM, Transformer, and ARIMA** models for high-accuracy predictions.
- Adept at bridging the gap between AI research and real-world applications.
- Excellent management, communication and people skills proven on **multi discipline teams** and running a consulting company.
- Accomplished Architect and experienced Team Lead with a strong track record.

Education

- **Professional Certificate in Machine Learning and Artificial Intelligence, UC Berkeley**
Notable Courses: Machine Learning Foundations, Deep Learning, NLP, Time Series Forecasting
- **Microsoft Certified Solution Developer (MCSD)**
- **Bachelor of Science: Applied Economics. University of San Francisco**
- **Computer Science, California State University - Chico (Transferred)**

Technology Stack

- **Languages/Tools:** Python, SQL, JavaScript, Docker, Git, FastAPI, AWS, RAG, LangChain
- **ML Frameworks:** PyTorch, Keras, TensorFlow, Scikit-learn, XGBoost, Pandas, NumPy
- **ML Techniques:** Time Series Forecasting, Feature Engineering, Regression, Transformers, LSTM, Natural Language Processing (NLP), Autoencoders, Clustering

Experience

ARNEY CONSULTING, INC – LEAD AI / MACHINE LEARNING ENGINEER/ARCHITECT **10/2024 – CURRENT**

Bitcoin Forecasting for LIVE Trading Automation

- Built a time series prediction pipeline using **SARIMAX, Lasso Regression**, and funding rate features
- Achieved **MAPE of 1.77%**, enabling high-confidence trade signals
- Processed **7M+ rows** of historical data and deployed via **Docker/AWS** for live forecasting
- Github: https://github.com/shawnarneygit/ai_machine

Deep Learning & NLP Experiments

- Implemented and tested **LSTM, GRU, Transformer** models on benchmark datasets
- Prototyped language modeling and sentiment analysis with HuggingFace transformers
- Applied **LangChain + RAG** for document-aware conversational AI systems

AI/ML Readiness: Collaborated with data analysts to prepare datasets for downstream analytics (Python scripts for CSV/JSON parsing).

Collaboration & Leadership: Led cross-functional teams to define data schemas and implement best practices around CI/CD (Docker, Git).

Data & Scalability: Architected a cloud-native system to manage millions of chemical records (petabytes of data).

- Designed RESTful APIs (Node.js, AWS Lambda) to ingest, index, and query large document stores.
- Built serverless ETL pipelines that normalize and store metadata in Elasticsearch for real-time search.

Tech Stack: JavaScript/ES6, Node.js, AWS (Lambda, S3, API Gateway), Docker, Elasticsearch, Electron, React.

Blockchain SDK Development: Architected and implemented iOS/Android/JavaScript SDKs for FIO Protocol (EOS fork).

- Developed core cryptographic routines in C++ and integrated them into mobile SDKs, demonstrating low-level algorithmic thinking.
- Published SDKs consumed by BRD, Coinomi, and Edge wallets—handled JSON parsing, API handshake, and error-handling logic.

Team Leadership: Mentored a team of four engineers on code reviews, unit testing, and agile best practices.

Data Integration: Created RESTful endpoints (Node.js) to interface with blockchain nodes; wrote scripts to ingest on-chain data for analytics.

Tech Stack: Swift, Kotlin, React Native, JavaScript, TypeScript, Node.js, C++, Android SDK, AWS (basic usage), Git.

Developed native iPhone and native Android apps for Digital Wallets (80 Credit Unions).

Technologies used: Objective-C, Swift, Python, Android SDK, Java.

Developed native iPad enterprise apps, Microsoft Windows metro app for in-branch credit union locations.

Technologies used: iPad-mini devices, beacons, card readers, Objective-C, Python (automating tasks), SOAP/xml, HTML/CSS/PhoneGap, Microsoft WCF/C# web services.

Manager, Architect, Mentor, Team Lead Developer for six mobile developers.

Mobile Banking apps for 90+ Credit Unions. Including: Wescom CU, Bayer FCU, Service 1st, Corning FCU, Baxter CU.

Open Source Code

AI/ML Projects

https://github.com/shawnarneygit/ai_machine/

FIO Protocol SDK's - a crypto blockchain based on the EOS coin.

https://github.com/fioprotocol/fiosdk_ios

https://github.com/fioprotocol/fiosdk_typescript